

CHAITANYA S.

Applied AI Engineer | Generative AI · RAG · Agentic AI · AI Platform Engineering

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PROFESSIONAL SUMMARY

Applied AI Engineer specializing in production-grade LLM applications, Retrieval-Augmented Generation (RAG), GraphRAG, agentic AI systems, and document-intelligence platforms built with Python, FastAPI, LangGraph, LangChain, LlamaIndex, vector databases, and LLM orchestration frameworks. Experienced designing grounded, citation-backed AI workflows using hybrid retrieval, reranking, structured outputs, evaluation, guardrails, and human-in-the-loop controls. Combines AI engineering with software-engineering experience in FDA-regulated pharmaceutical environments, including 21 CFR Part 11, GxP, and ALCOA+, to build secure, auditable, compliance-aware AI systems.

CORE SKILLS

Generative AI & LLMs: RAG Architecture, LangChain, LlamaIndex, OpenAI API, GPT-4/4o, Hugging Face Transformers, Prompt Engineering, Structured Outputs, Hallucination Mitigation

Agentic AI & Automation: AI Agents, LangGraph, Multi-Agent Systems, Model Context Protocol (MCP), Tool / Function Calling, Task Routing, Human-in-the-Loop Approvals, Sandboxed Execution

Retrieval & NLP: Embeddings, Semantic Search, Semantic Chunking, Hybrid Search, BM25, GraphRAG, Dense Vector Retrieval, Metadata Filtering, Reranking, Citation-Grounded Generation

AI Evaluation & Safety: Phoenix, LangSmith, Groundedness Evaluation, Retrieval Precision, NeMo Guardrails, Llama Guard, Prompt-Injection Defense, Audit Logging

Backend & Infrastructure: Python, SQL, FastAPI, REST APIs, Async Architecture, Pydantic, Streamlit, Pandas, NumPy, PostgreSQL, ChromaDB, Pinecone, Docker, AWS EC2/S3, CI/CD, Git, GitHub

Security & Compliance: RBAC, FDA 21 CFR Part 11, GxP, ALCOA+, Regulatory Document Workflows, SOP, CAPA, Deviation Workflows, Validation-Oriented Systems

PROFESSIONAL EXPERIENCE

Founder & Applied AI Engineer

June 2026 - Present

SPACTR AI Labs LLC - Milwaukee, WI (Remote)

Tech Stack: Python, FastAPI, LangGraph, MCP, LangChain, LlamaIndex, ChromaDB, Pinecone, NeMo Guardrails, Llama Guard, Streamlit, Docker, AWS EC2/S3, OpenAI API, Hugging Face

Enterprise AI Platforms Developed:

PharmaAI Platform - Regulatory, Summarization & Compliance Intelligence System

- Architected an end-to-end RAG pipeline using LlamaIndex and custom ingestion workflows covering semantic chunking, vector indexing, reranking, and citation generation for FDA guidance documents and regulatory PDFs.
- Extended retrieval precision using hybrid search, dense vectors, BM25, and GraphRAG for relationship-aware query resolution across regulatory, SOP, CAPA, and deviation workflows, evaluating groundedness and latency through Phoenix, LangSmith, and custom evaluation suites.
- Engineered a compliance-focused QA assistant integrating NeMo Guardrails, Llama Guard, prompt-injection defense, structured outputs, mandatory inline citations, and evidence-backed answer generation.
- Consolidated retrieval, summarization, compliance QA, and document-intelligence workflows behind shared asynchronous FastAPI services with RBAC, audit logging, and validation review controls.

Agentic AI Orchestration Platform - Local-First & Private-Cloud Multi-Agent System

- Built a local-first, privacy-focused multi-agent orchestration platform on LangGraph using deterministic state-machine control, modular agents, dynamic task routing, and execution planning.
- Implemented tool/function calling with human-in-the-loop approval gates and sandboxed execution runtimes so operators explicitly authorize sensitive AI actions before execution.
- Integrated external tools through Model Context Protocol (MCP) servers, establishing a standardized interface for agents to query local files, PostgreSQL databases, REST APIs, and external systems.
- Developed a database-backed wait-state pattern that serializes agent state and memory to PostgreSQL during approval pauses and resumes execution without context loss.

AI/ML Engineer

May 2024 - May 2026

SolutionsMax Technology Services Inc - Sacramento, CA (Remote)

Tech Stack: Python, FastAPI, OpenAI API, LangChain, Pinecone, PostgreSQL, Docker, AWS EC2/S3, REST APIs, CI/CD

- Shipped production LLM-powered summarization, entity extraction, and knowledge-retrieval features for enterprise regulatory documentation, processing 50,000+ pages of compliance text.
- Built retrieval-based AI capabilities into enterprise backend services through RESTful FastAPI microservices, maintaining sub-300 ms API response latency under high-volume query loads.
- Designed and optimized vector-search indexing pipelines using Pinecone and PostgreSQL, including custom chunking, embedding workflows, and metadata filtering to improve retrieval accuracy.
- Implemented prompt-engineering patterns, Pydantic-based structured JSON outputs, and validation guardrails to reduce hallucination risk across automated compliance workflows.
- Built preprocessing and validation workflows that converted unstructured regulatory content into retrieval-ready chunks and Pydantic-validated structured outputs for downstream AI services.
- Containerized AI microservices using Docker and deployed scalable API endpoints on AWS EC2 and S3 with CI/CD practices for repeatable releases.
- Partnered cross-functionally with product and engineering teams to design automated document-intelligence workflows, reducing manual regulatory review cycles by 35%.

Software Engineer - Regulated Data Systems

June 2020 - July 2023

SolutionsMax Technology Services Pvt Ltd - Visakhapatnam, India

Tech Stack: Python, SQL, PostgreSQL, ETL Pipelines, Data Modeling, QA Automation

- Engineered Python and SQL data and reporting pipelines for FDA-regulated systems aligned with 21 CFR Part 11 and ALCOA+ data-integrity standards.
- Designed relational data models and automated ETL workflows that improved data traceability and auditability across pharmaceutical product-lifecycle systems.
- Built validation, transformation, and reporting workflows for structured enterprise datasets supporting regulatory, quality, and compliance processes.
- Automated validation and reporting processes, reducing manual QA testing overhead by 40% and strengthening audit readiness across regulated datasets.
- Authored functional specifications and technical design documents mapping regulatory mandates and business requirements into implementation-ready engineering deliverables.

EDUCATION

M.S., Computer Science / Information Systems

Dec 2025

Concordia University Wisconsin - USA

B.S., Computer Science

Jun 2020

GITAM University - India